

Airframe Technology Master

Certification

Lecture 03

Structural Requirements (I)

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April 2025

STRUCTURAL REQUIREMENTS (I)

AGENDA

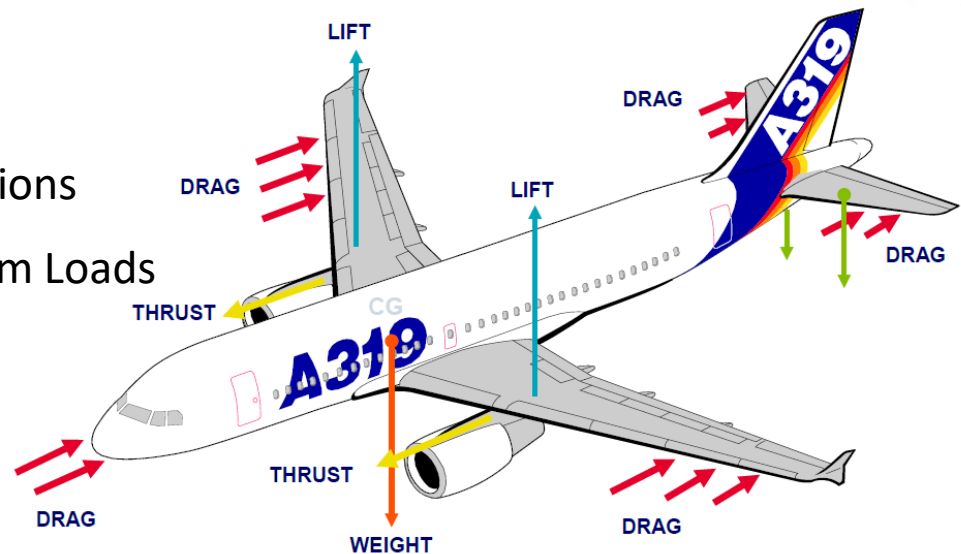
- Applicable structural requirements - Loads
 - Flight Maneuvers, gust and turbulence loads.
 - Pressurised compartment loads
 - Ground loads
- Applicable structural requirements – Static Strength
 - CS 25.301 Loads
 - CS 25.303 Factor of safety
 - CS 25.305 Strength and deformation
 - CS 25.307 Proof of structure
 - CS25.963 & CS 25.965 Fuel tanks

AGENDA

- Applicable structural requirements – Materials
 - CS25.601 General Design and Construction
 - CS 25.603 Materials
 - CS25.605 Fabrications Methods
 - CS25.609 Protection of Structure
 - CS25.613 Material strength properties and Material Design Values
 - CS25.619 Special factors
 - CS 25.621 Casting factors
 - CS 25.623 Bearing factors
 - CS 25.625 Fitting factors
- Applicable structural requirements – Composite (ATA029)
- Airframe Documentation process

Flight Manoeuvres Loads

- CS 25.331 Symmetric manoeuvring conditions
- CS 25.333 Flight manoeuvring envelope
- CS 25.335 Design airspeeds
- CS 25.337 Limit manoeuvring load factors
- CS 25.341 Gust and turbulence loads
- CS 25.345 High lift devices
- CS 25.349 Rolling conditions
- CS 25.351 Yaw manoeuvre conditions
- CS 25.395 & 25.397 Control System Loads



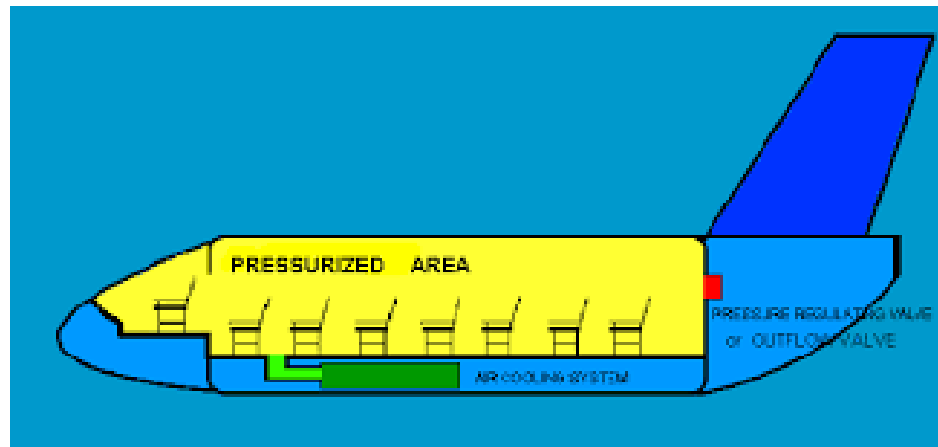
CS 25.365 Pressurised compartment loads

For aeroplanes with one or more pressurised compartments the following apply:

- (a) The aeroplane structure must be strong enough to **withstand the flight loads combined with pressure differential loads** from zero up to the maximum relief valve setting.
- (b) The external pressure distribution in flight, and stress concentrations and fatigue effects must be accounted for.
- (c) If landings may be made with the compartment pressurised, landing loads must be combined with pressure differential loads from zero up to the maximum allowed during landing.
- (d) The aeroplane structure must be strong enough to withstand the pressure differential loads corresponding to the maximum relief valve setting multiplied by a factor of 1.33, omitting other loads.

CS 25.365 Pressurised compartment loads

(e) Any structure, component or part, inside or outside a pressurised compartment, the failure of which could interfere with continued safe flight and landing, must be designed to **withstand the effects of a sudden release of pressure** through an opening in any compartment at any operating altitude [...]



Ground loads

- CS 25.479 Level landing conditions
- CS 25.481 Tail-down landing conditions
- CS 25.483 One-gear landing conditions
- CS 25.485 Side load conditions
- CS 25.487 Rebound landing condition
- CS 25.489 Ground handling conditions
- CS 25.491 Taxi, takeoff and landing roll
- CS 25.493 Braked roll conditions
- CS 25.495 Turning
- CS 25.497 Tail-wheel yawing
- CS 25.499 Nose-wheel yaw and steering
- CS 25.503 Pivoting
- CS 25.507 Reversed braking
- CS25.509 Towing loads



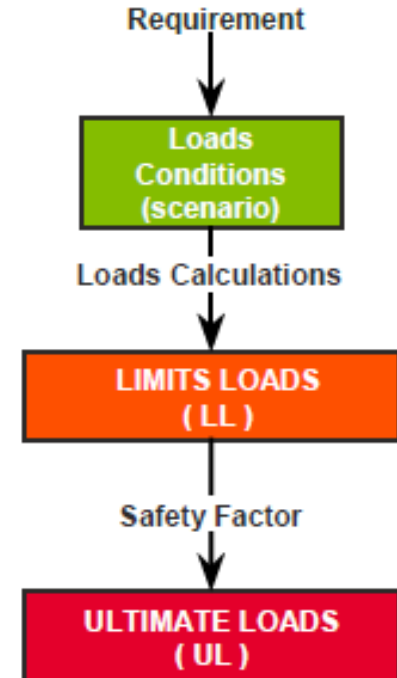
CS 25.301 Loads

- (a) Strength requirements are specified **in terms of limit loads** (the maximum loads to be expected in service) and ultimate loads (limit loads multiplied by prescribed factors of safety). **Unless otherwise provided, prescribed loads are limit loads.**
- (b) Unless otherwise provided the specified air, ground, and water loads must be placed in equilibrium with inertia forces, considering each item of mass in the aeroplane. These loads must be distributed to conservatively approximate or closely represent actual conditions. (See AMC No. 1 to CS 25.301(b).) Methods used to determine load intensities and distribution must be **validated by flight load measurement** unless the methods used for determining those loading conditions are shown to be reliable. (See AMC No. 2 to CS 25.301(b).)
- (c) If **deflections under load** would significantly change the distribution of external or internal loads, this redistribution must be taken into account.

CS 25.303 Factor of Safety

Unless otherwise specified, a factor of safety of 1.5 must be applied to the prescribed limit load which are considered external loads on the structure. When loading condition is prescribed in terms of ultimate loads, a factor of safety need not be applied unless otherwise specified.

- Factor of safety is the mandatory margin to cover the variability on:
 - Loads / abnormal loads
 - Production / workmanship
 - Analysis
 - Materials properties
 - Thickness and strength degradation during service life



CS 25.305 Strength and deformation

- (a) The structure must be able to **support limit loads without detrimental permanent deformation...**
- (b) The structure must be able to **support ultimate loads without failure for at least 3 seconds...** Static tests conducted to ultimate load must include the ultimate deflections and ultimate deformation induced by the loading...
- (c) Where structural flexibility is such that any rate of load application likely to occur in the operating conditions might produce transient stresses appreciably higher than those corresponding to static loads, **the effects of this rate of application must be considered.**

CS 25.307 Proof of structure

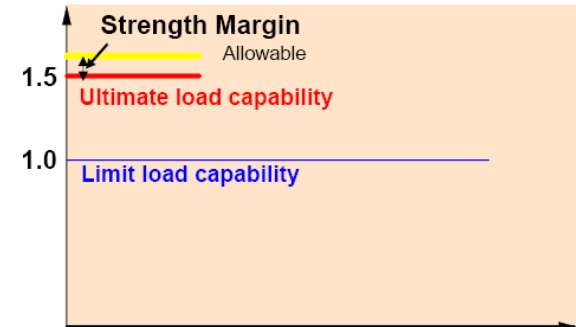
- (a) Compliance with the strength and deformation requirements of this Subpart must be shown for each critical loading condition. **Structural analysis may be used only if the structure conforms to that for which experience has shown this method to be reliable. In other cases, substantiating tests must be made** to load levels that are sufficient to verify structural behaviour up to loads specified in CS 25.305
- Based on this requirement general approach for structure is to demonstrate compliance with the airworthiness requirements by analysis supported by test (MoC 2, 4). MoC 4 (testing) as only compliance demonstration is only used in exceptional cases.

CS 25.307 Proof of structure (Cont'd)

- For structure under compression loading, occurrence of buckling should be verified.



- Reserve Factor Limit= Allowable L_L / Limit ≥ 1.00
- Reserve Factor Ultimate= Allowable U_L / Ultimate ≥ 1.00



Classification of structure (AMC 25.307)

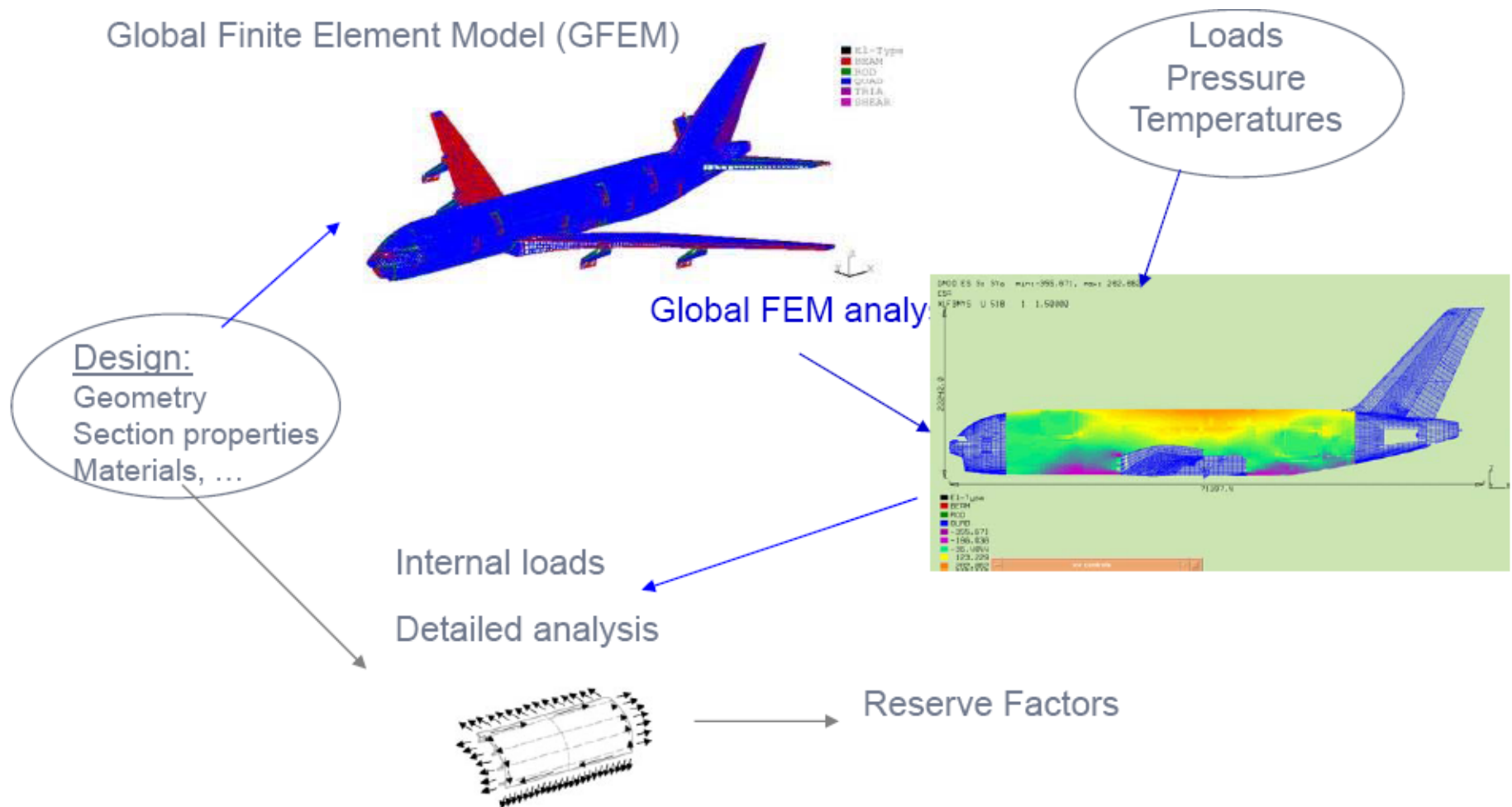
	Identification	Definition
N	New structure	Structure for which behaviour is not adequately predicted by analysis supported by previous test evidence. Structure that utilises significantly different structural design concepts such as details, geometry, structural arrangements, and load paths or materials from previously tested designs.
SN	Similar new structure	Structure that utilises similar or comparable structural design concepts such as details, geometry, structural arrangements, and load paths and materials to an existing tested design.
S	Derivative/similar structure	Structure that uses structural design concepts such as details, geometry, structural arrangements, and load paths, stress levels and materials that are nearly identical to those on which the stressing methods have been validated.

Classification of structure is key to define the required testing:

- For “New” structures the MoC is Analysis, supported by new strength testing of the structure to Limit and Ultimate Loads (typically at full scale level).
- For “Similar/New” structures, MoC is by Analysis, validated by previous test evidence supported by additional limited testing as required (component level).
- For “Derivative/Similar” structures, MoC is by Analysis

CS 25.307 Proof of structure (Cont'd)

Structure analysis:



CS 25.307 Proof of structure (Cont'd)

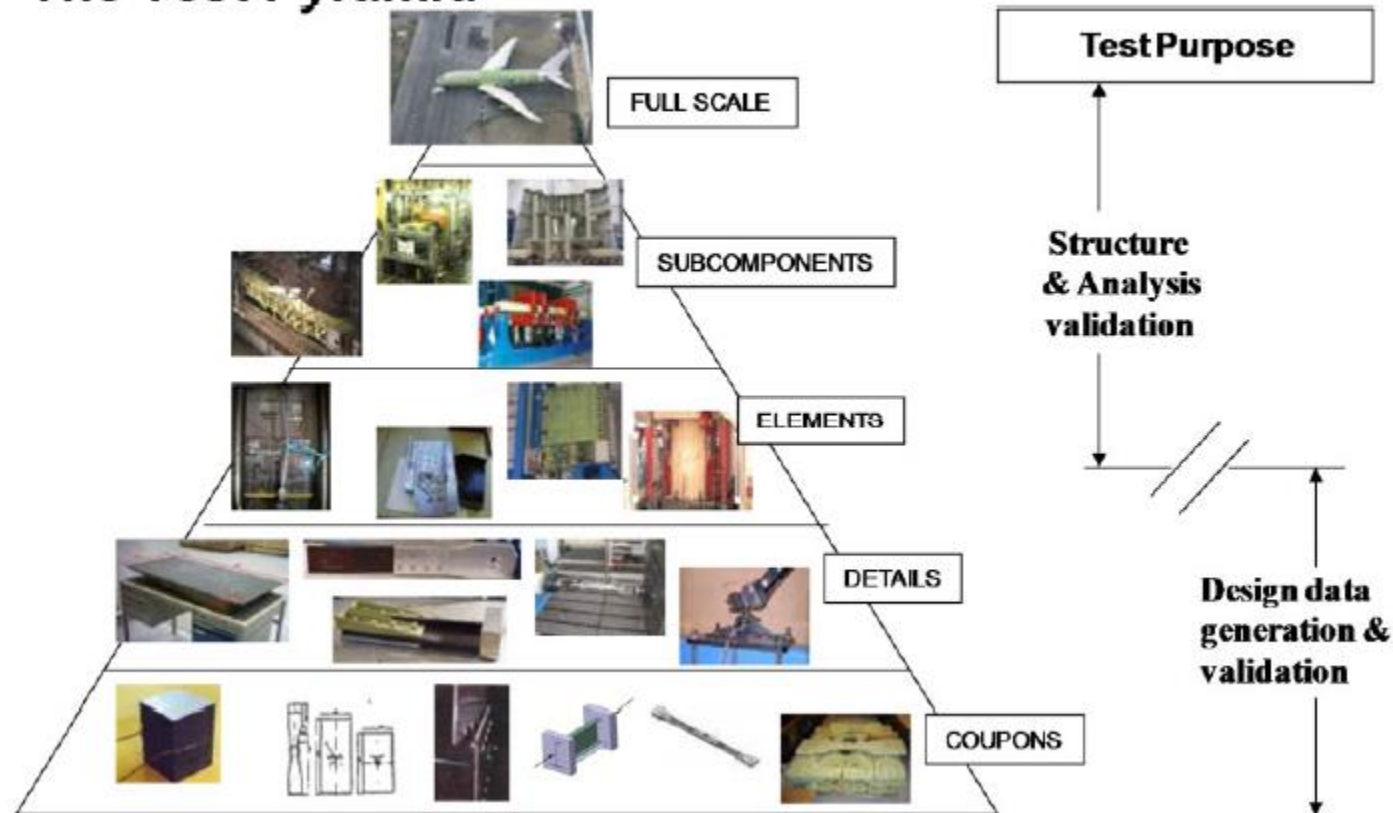
- Analysis validation:



CS 25.307 Proof of structure (Cont'd)

- Analysis validated by testing approach leads to detailed definition of required tests at different levels:

=> The Test Pyramid



Design and Construction

CS 25.601 General

The aeroplane may not have design features or details that experience has shown to be hazardous or **unreliable**. The suitability of each questionable design detail and part must be **established by tests**.

CS 25.609 Protection of structure

Each part of the structure must -

(a) Be **suitably protected** against deterioration or loss of strength in service due to any cause, including –

- (1) **Weathering;**
- (2) **Corrosion; and**
- (3) **Abrasion; and**

(b) Have provisions for ventilation and drainage where necessary for protection.

CS 25.603/ CS 25.605/ CS 25.613

CS 25.603 Materials (For Composite Materials, see AMC 20-29)

The **suitability and durability of materials** used for parts, the failure of which could adversely affect safety, must –

- (a) Be established on the basis of **experience or tests**;
- (b) Conform to approved **specifications**, that ensure their having the strength and other properties assumed in the design data; and
- (c) Take into account the effects of **environmental** conditions, such as temperature and humidity, expected in service.

CS 25.605 Fabrication methods

- (a) The methods of fabrication used must produce a consistently sound structure
- (b) Each new aircraft fabrication method must be substantiated by a test programme.

CS 25.613 Material strength properties and Material Design Values

- (a) Material strength properties must be based on enough tests of material meeting approved specifications **to establish design values on a statistical basis.**
- (b) Material design values must be chosen to minimise the probability of structural failures due to material variability. Except as provided in sub-paragraphs (e) and (f) of this paragraph, compliance must be shown by selecting material design values which assure material strength with the following probability:
- (1) Where applied loads are eventually distributed through a **single member** within an assembly, the failure of which would result in loss of structural integrity of the component, **99% probability with 95% confidence.**
 - (2) For **redundant structure**, in which the failure of individual elements would result in applied loads being safely distributed to other load carrying members, **90% probability with 95% confidence.**
- (c) The **effects of environmental conditions**, such as temperature and moisture, on material design values used in an essential component or structure **must be considered** where these effects are significant within the aeroplane operating envelope. [...]

CS 25.963 & CS 25.965 Fuel tanks

CS 25.963 - Fuel tanks: general

- (a) Each fuel tank must be able to withstand, without failure, the **vibration, inertia, fluid and structural loads** that it may be subjected to in operation.

CS 25.965 – Fuel tank tests

- (a) It must be shown by tests that the fuel tanks, as mounted in the aeroplane can withstand, without failure or leakage, the more critical of the **pressures** resulting from the conditions specified in the sub-paragraphs (a)(1) and (2) of this paragraph. In addition it must be shown by either analysis or tests, (see AMC 25.965(a)) that tank surfaces subjected to more critical pressures resulting from the conditions of sub-paragraphs (a)(3) and (4) of this paragraph are able to withstand the following pressures:

- (1) An **internal pressure** of 24 kPa (3.5 psi)
- (2) 125% of the maximum air pressure developed in the tank from **ram effect**.
- (3) **Fluid pressures** developed during **maximum limit accelerations**, and deflections, of the aeroplane with a full tank.
- (4) **Fluid pressures** developed during the most adverse combination of **aeroplane roll** and fuel load.

CS 25.619 Special factors

The factor of safety prescribed in CS 25.303 must be multiplied by the highest pertinent special factor of safety prescribed in CS 25.621 through CS 25.625 for each part of the structure whose strength is –

- (a) Uncertain.
- (b) Likely to deteriorate in service before normal replacement; or
- (c) Subject to appreciable variability because of uncertainties in manufacturing processes or inspection methods.

- **Special factors (Cont'd)**
- CS 25.621 Casting factors (up to 2.0)
- CS 25.623 Bearing factors (part that has clearance -free fit-, and is subject to pounding or vibration)
- CS 25.625 Fitting factors
 - (a) For each fitting whose **strength is not proven by limit and ultimate load tests** in which actual stress conditions are simulated in the fitting and surrounding structures, a **fitting factor of at least 1.15** must be applied to each part of –
 - (1) The fitting; (2) The means of attachment; and (3) The bearing on the joined members. [...]

Typical structure MoCs

Most used MoC

MOC	Title
0	Description/ statement
1	Design review
2	Analysis
4	Laboratory Test

For some specific aspects

MOC	Title
5	Ground tests on A/C
6	Flight tests

Calculation and stress analysis :

- Static strength
- Fatigue strength
- Damage tolerance strength

Static tests :

- up to ultimate loads for nominal structures and up to limit load including artificial damages

Fatigue tests :

- Damage initiation and initial crack propagation

Damage tolerance tests :

- Observation of natural and artificial crack propagation/residual strength

Tests on complete aircraft:

- Internal pressure, ground vibration, functioning under deformation and flight test, loads validation

CS Paragraphs most affected in structural mods

Materials

CS26.613

Properties A, B and typical

CS25.603

Composite material

CS25.603

Fabrication Methods

CS25.605

Static Strength

25.305

25.307

Bird strike

25.631

Fatigue and DT

CS25.571

Design and construction

25.601

25.619

25.621

25.623

25.625

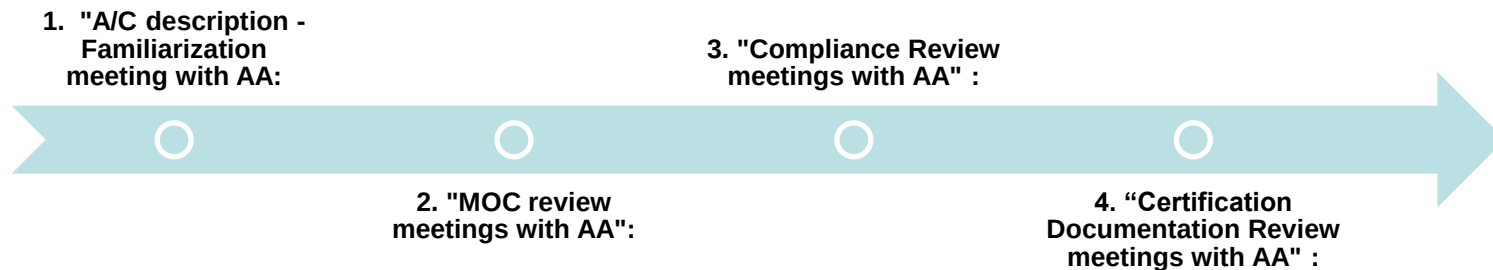
Structure protection

25.609

DOCUMENTATION PROCESS- MAIN STEPS

As part of the demonstration to Airworthiness Authority of conformance, **structural compliance regarding Design, Static Strength, Fatigue Strength, Damage Tolerance, and particular risk will be covered by a series of Airframe Certification Documents.**

It is necessary coordinate the compliance demonstration process with progressive steps of showing the Authority the compliance by different reviews and final documentation delivery:



The documentation structure should reflect this process

PRESENTATION TO AAs

Typically 3 meetings for a certification project:

Meeting 1 – Familiarization meeting.

Main A/C design characteristics per section

Meeting 2 – MoC meeting

Proposed MoC per section to show compliance with the applicable requirements

Meeting 3 – Results/Compliance meeting

Detailed review of all previous agreed MoC.

Overview of critical items, critical results and summary of test results

Typical volumes of Airframe Certification Documentation

- Structural Description
- Methods for compliance demonstration
- Static Strength Justification
- Fatigue and Damage Tolerance Justification
- Comparison of Static test results with analysis
- Comparison fatigue and damage tolerance test results with analysis
- Demonstration of particular risks (Bird impact, rotor burst...)

Structural description

- Documentation includes a **general description** of the structure with structural **geometrical data**.
- The data shall include **pictures** and detailed tables specifying relevant geometry ratios and all the relevant design features of the structure.
- **Structure categorization as primary-secondary.**
- **Materials and manufacturing processes**
- **Design requirements**

Methods for compliance demonstration

The data included in this volume shall comprehensively cover the method of conformity demonstration. The data shall include:

- A short description of the **main A/C sections**
- The classification of the A/C sections, i.e. **Similar, Similar New or New**
- The MoC certification methods for justifying the structure including test approach

Static Strength Justification

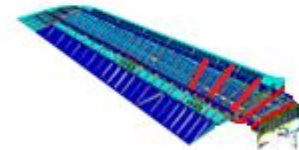
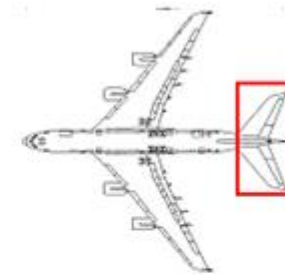
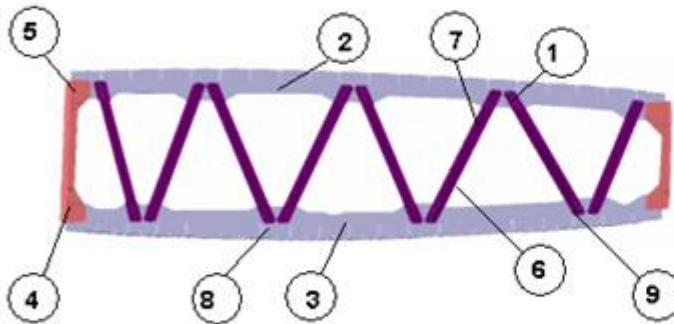
This volume shall consist of pages containing the **detailed tables of reserve factors** and **picture sheets showing the critical location** of the entry.

All entries shall be selected from the strength summary produced as an integral part of the check stress and can be selected in accordance with the following criteria:. .

- **All structural components.**
- Sufficient entries to demonstrate **all modes of failure** have been considered.
- **Sufficient entries to provide** a clear picture of the progress of the stress and **margin envelopes** over the structural component.
- Due account shall be taken of similar reserve factors resulting from **different load case**

Static Strength Justification (Example 1/2)

Rib



STRUCTURAL JUSTIFICATION - STATIC LOADS

AIRBUS

Static Strength Justification (Example 2/2)

Kind of Loading: S = Shear, T = Tension, C = Compression, B = Bearing, Bn = Bending							RF = $\frac{\text{Allowable value}}{\text{Ultimate value}}$		
Nr	Location	Material	Load Case	Strain Stress Force	Ultimate Value	Kind of Loadi ng	Allowable Value	RF	Remarks
				Str St F	Percent MPa N		Percent MPa N		
1	RIB								
2									
3									
4									
5									
6									
7									
8									
9									

Basic Report:

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STRUCTURAL JUSTIFICATION - STATIC LOADS

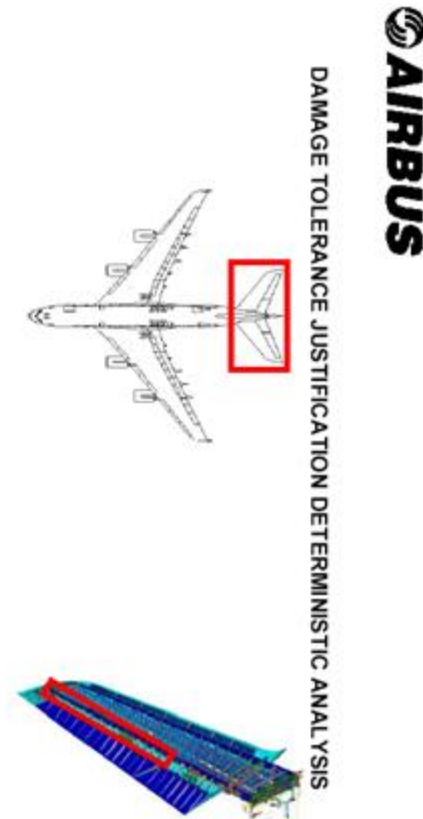
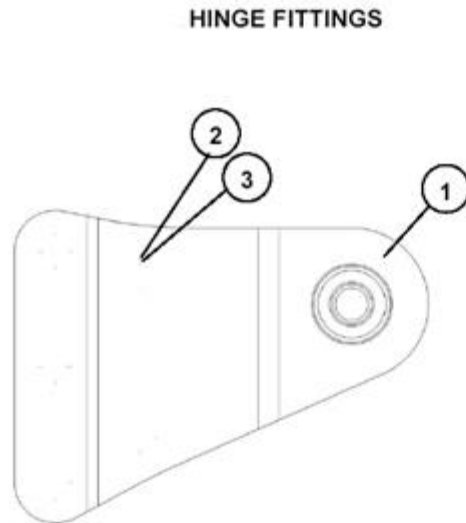
Fatigue Damage Tolerance Justification

The contents of this document will be based on extracts from the Fatigue and Damage Tolerance (F&DT) check stress.

Selection criteria include:

1. The fatigue live factors for various features .
2. The crack propagation lives and associated data for the location .

Fatigue and DT Justification (Example 1/2)



Fatigue and DT Justification (Example 1/2)

No	LOCATION	MATERIAL	FATIGUE		CRACK PROPAGATION					RESIDUAL STRENGTH		REMARKS
			Unfactored Fatigue Life (flights)	LF _{DSG}	Crack Length			Unfactored Crack Growth Life		Stress or Load	Load Case	
					Equivalent manufact. defect a ₀ (mm)	Detectable a _{det} (mm)	Critical a _{crit} (mm)	a ₀ to a _{det} (flights)	a _{det} to a _{crit} (flights)			
1	HINGE FITTING LUGS											
2	HINGE FITTING											
3	HINGE FITTING JOINT											

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Basic Report: ...

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DAMAGE TOLERANCE JUSTIFICATION DETERMINISTIC ANALYSIS



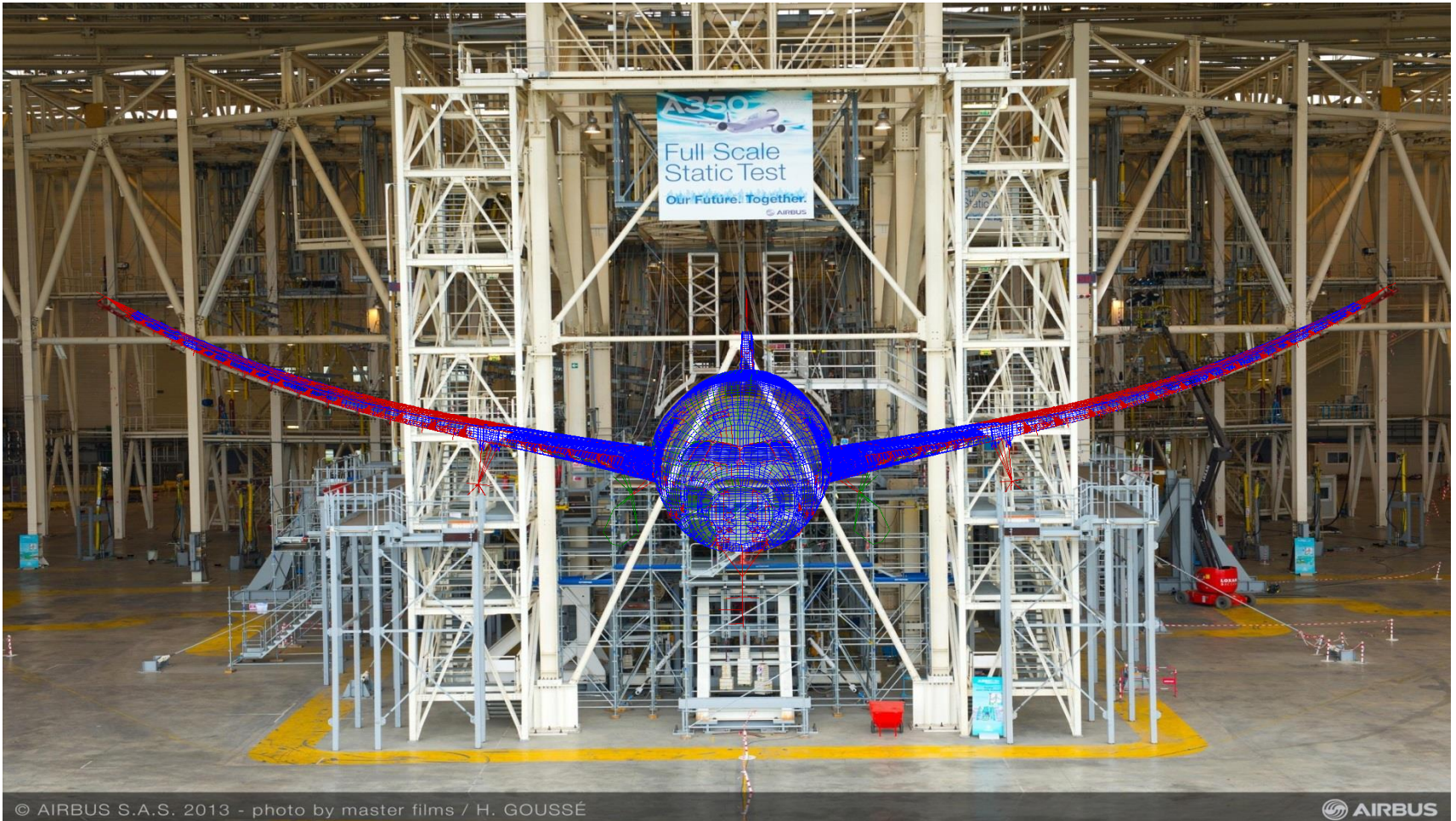
Comparison of Static test results with analysis

This document summarizes the **static tests** which have been or will be carried out for the aircraft certification with the objective of **validating the calculation methods** applied for the justification of all structural items.

The summary shall contain:

- Summary of test data, load cases and load factors reached, test results
- Conclusion of test (for instance introduction of modifications)
- Comparison of the test results and the predictions

Comparison of Static test results with analysis – Test Correlation



CS 25.603 Materials

CS 25.603 – Materials

The **suitability and durability** of materials used for parts, the failure of which could adversely affect safety, must –

- (a) **Be established** on the basis of experience or tests;
- (b) Conform to approved **specifications**, that ensure their having the strength and other properties assumed in the design data (See AMC 25.603(b); and
- (c) Take into account **the effects of environmental conditions**, such as temperature and humidity, expected in service.

Nothing specific to composite material

COMPOSITES IN CS 25

Specific Requirements for Composite Aircraft Structure

- Up to **CS25 Amendment 8**, 18 December 2009
 - CS 25.603 Materials (For Composite Materials see **AMC No. 1 and No. 2 to 25.603.**)
- **CS25 Amendment 9**, 5 August 2010 Includes Composite with new AMC 20-29,
 - CS 25.603 Materials (For Composite Materials, see **AMC 20-29**)



AMC 20-29



AC 20-107B

COMPOSITES IN CS 25

- **CS25 Amendment 9**, 5 August 2010 Includes Composite with new AMC 20-29,
— CS 25.603 Materials (For Composite Materials, see **AMC 20-29**)

The final mechanical behaviour of a given composite material may vary greatly depending on the **processing methods** employed to fabricate production parts.

Special care needs to be taken in controlling both the materials being procured and how the material is processed once delivered to the fabrication facility

... a need to protect structures against the **degradation** possible in service.

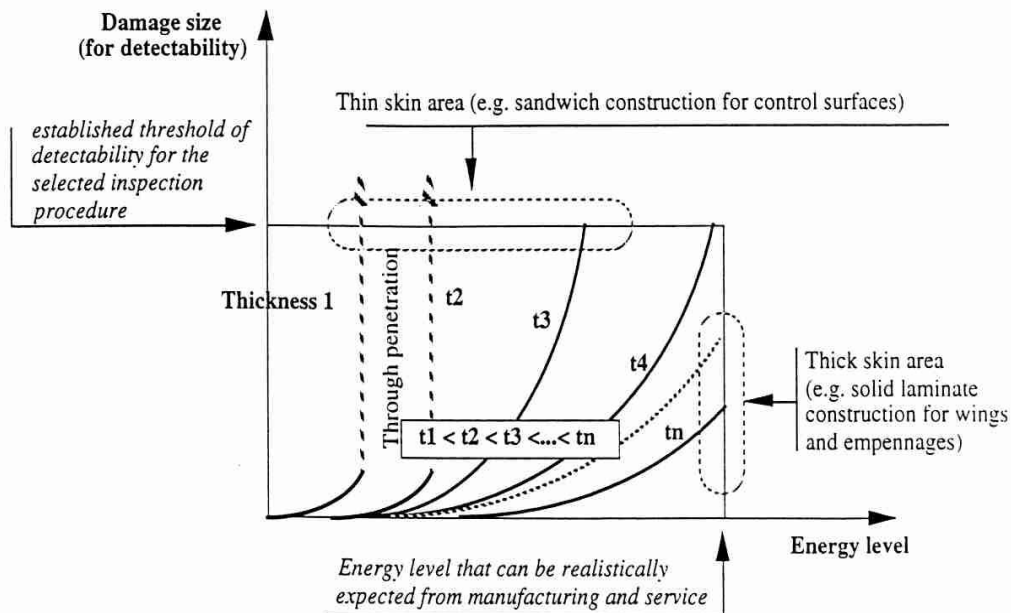
They also require that the design account for any changes in performance (e.g., **environmental and variability effects**) permitted by material and process specifications.

ENVIRONMENTAL EFFECTS

1. Temperature
 - Ambient
 - Solar radiation effects/Solar heating effect
 - Absorvivity values depending on colour
 - Cooling effect due to taxi, take-off and climb
2. Moisture uptake:
 - Humidity (85%RH)
 - Temperature (70deg C)
 - Laminate thickness
3. Bonding solutions
4. Aggressive fuels (skydrol, fuel, de-icing)
5. UV radiation effects on resin.
6. Tg (Glass transition) versus operating envelope
7. Water ingress
8. Flutter (weight increase due to water absorption, stiffness decrease)

IMPACT DAMAGE

It should be shown that impact damage that can be expected from **manufacturing and service**, but not more than the established threshold of detectability for the selected inspection procedure, will not reduce the structural strength **below ultimate load** capability.



IMPACT DAMAGE – BVID AND VID

When using a **visual inspection** procedure, the likely impact damage at the threshold of reliable detection has been called **barely visible impact damage (BVID)**. Selection of impact sites for static strength substantiation should consider the **criticality** of the local structural detail, and the ability to inspect a location.

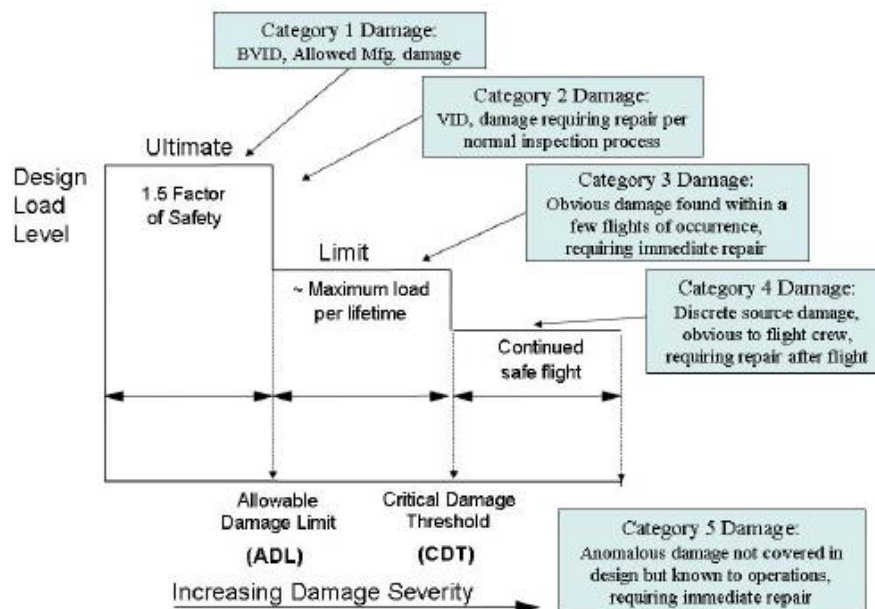
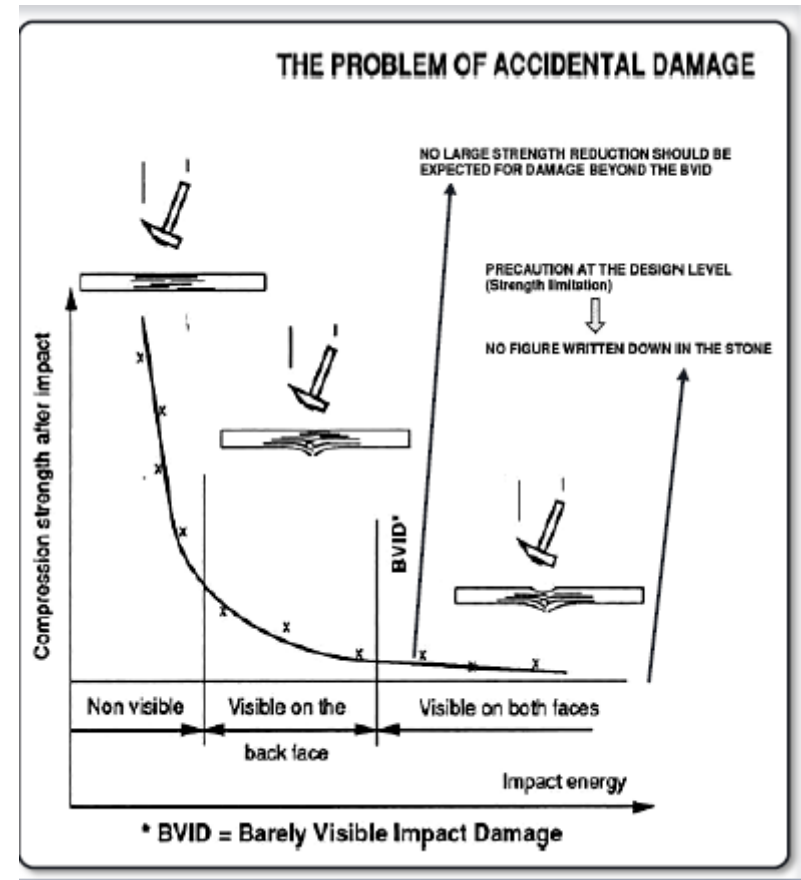


Figure 3 - Schematic diagram showing design load levels versus categories of damage severity.

ACCIDENTAL IMPACT DAMAGES

1. Low Velocity (drop of tools)
2. High velocity (runway debris)
3. Static strength reduction in compression
4. Substantiation by limitation of strains and non-growth approach
5. BVID – UL after fatigue
6. VID – LL after non-growth
7. Threshold of detectability: dent depth
8. Energy cut-off: manufacturing, maintenance, in-service



PROOF OF STRUCTURE – STATIC STRENGTH

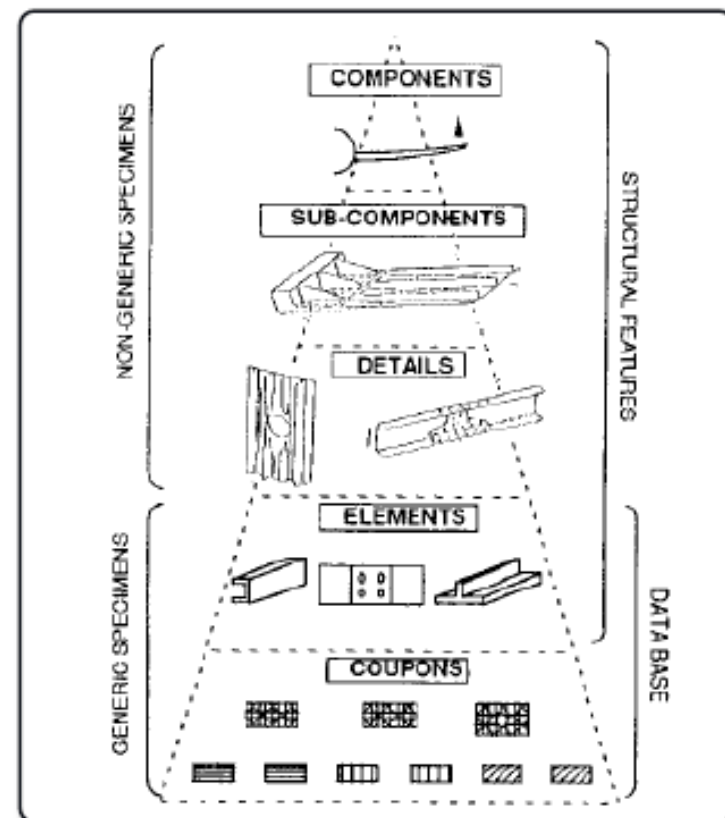
1. Environmental effect considered as Knock Down Factors KDF in analyses and test
2. Ultimate strength after fatigue
3. Accidental impact damage
4. Minimum Quality test article (Manufacturing defects)
5. Thermal stresses
6. Material B-values
7. Repair and allowable damages

PROOF OF STRUCTURE – FATIGUE AND DAMAGE TOLERANCE

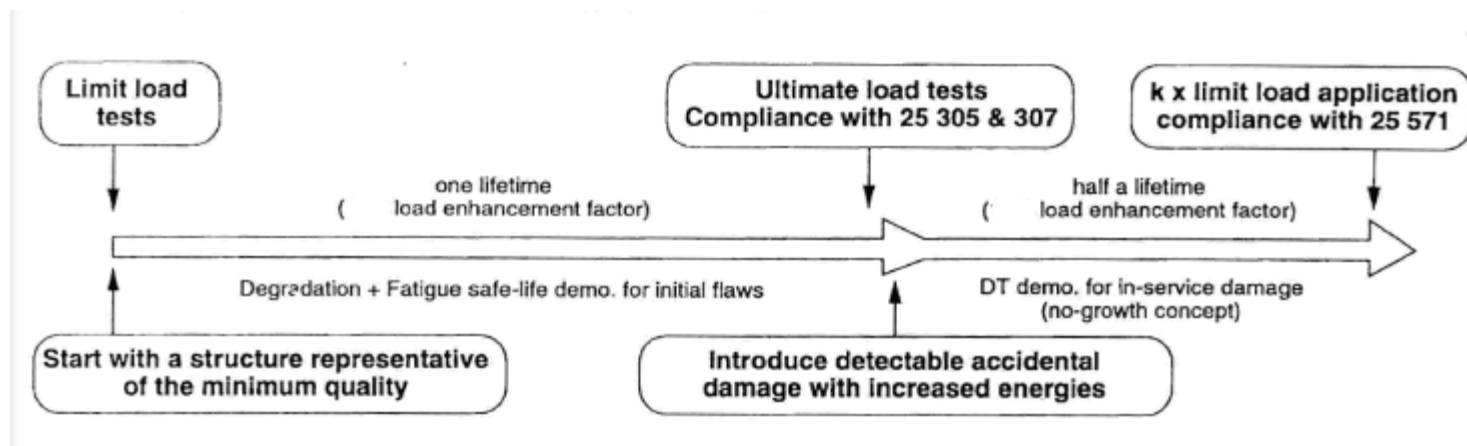
1. Fatigue insensitivity – 3D details
2. Non detrimental growth of impacts and damages
3. Impact damages – Compression after impact criterion
4. Residual strength for large damage capability assessment
5. Bonded elements – LL capability with disbonds
6. Lightning strike damage
7. Repair and allowable damages

PROOF OF STRUCTURE – TESTS

- Composite DT and fatigue evaluations require substantiation in **component tests** unless experience with similar designs, material systems, and loadings is available to demonstrate the adequacy of the analysis supported by coupons, elements, and sub-component tests.
- Final **static strength, fatigue, and damage tolerance** substantiation may be gained in testing a single component test article if sufficient **building block test evidence exists** to ensure that the selected sequence of repeated and static loading yield results representative of that possible in service or provide a conservative evaluation.



COMPONENT TEST – TEST PHASES



Life (or Load) Enhancement Factor: An additional load factor and/or test duration applied to structural repeated load tests, relative to the intended design load and life values, used to account for material variability. It is used to develop the required level of confidence in data.

Advantages and challenges of composite materials

Advantages:

- High Static Strength and Stiffness
- Optimum design → Anisotropic material
- Linear elastic behaviour till rupture
- Thermal expansion coefficient close to 0
- Excellent Fatigue strength behaviour
- No corrosion

Challenges:

- Material characteristics variation due to production
- No plastic deformation
- Effect of impacts
- Internal stress when combined with metal
- Water absorption, temperature and UV sensibility
- Out of plane characteristics (peel, offsets, eccentricities)
- Bad electrical conductivity